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CSA0503-DATABASE MANAGEMENT SYSTEMS

Strong vs Weak Entities

in ER Modeling

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What Is an Entity?

A building block of the ER model that represents a real-world object or concept



Entity

A distinguishable real-world object — a person, place, concept, or event (e.g., STUDENT, BRANCH).



Entity Set

A collection of similar entities that share the same attributes (e.g., all STUDENT records).



Attribute

A property that describes an entity (e.g., Name, DOB, Roll_No).



Key

One or more attributes that uniquely identify each entity instance in the set.

Not every entity is the same kind of building block

How to identify entity as strong vs weak?

Every entity in a schema answers this test

“Can this entity be uniquely identified using only its own attributes—with no reference to any other entity?”

YES

→ STRONG ENTITY

It owns a complete primary key and exists independently in the database.

NO

→ WEAK ENTITY

It needs a partial key plus its owner entity's key to be uniquely identified.

STRONG ENTITY

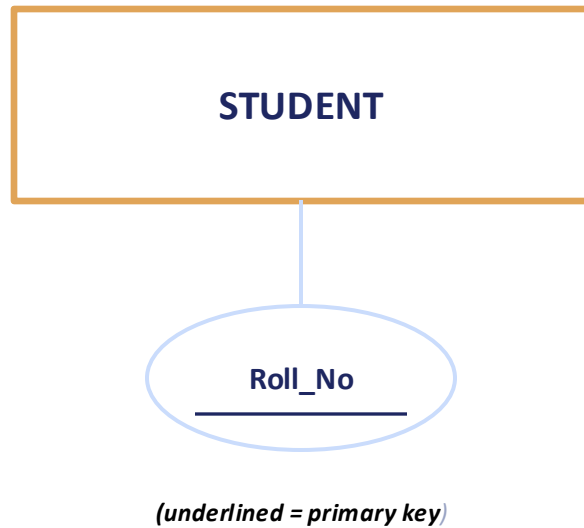
Definition & Characteristics

A strong entity is one that has a complete primary key of its own and can be uniquely identified without depending on any other entity.

- Owns a full primary key (e.g., Employee_ID, Roll_No, ISBN) that is unique across the entire entity set
- Existence in the database does not rely on any other entity being present
- Can be inserted, updated, or deleted independently of other tables
- Typically forms the 'one' side of a one-to-many relationship with a dependent (weak) entity
- Also called a 'regular entity' or 'parent entity' in some textbooks

STRONG ENTITY · ER NOTATION

Diagram



Notation Rules

- Single-line rectangle = the entity itself
- Single-line oval, underlined text = the primary key attribute
- A regular (single) diamond connects it to related entities
- No dependency arrows or double borders are needed

Banking Case Study: EMPLOYEE

Schema

EMPLOYEE (Employee_ID, Name, Designation, Branch_ID)

Employee_ID is the primary key — no other table's data is needed to uniquely pick out one employee.

Referenced by:

- LOAN.Approved_By → Employee_ID
- ACCOUNT.Managed_By → Employee_ID

Why It's Strong

- Employee_ID alone, with no help from Branch, Loan, or Account, identifies every row uniquely
- An employee record can be created the moment someone joins — before any account or loan exists
- Deleting a Loan or Account record has no effect on the Employee row
- In the ER diagram, EMPLOYEE is the 'owner' side of every relationship it takes part in

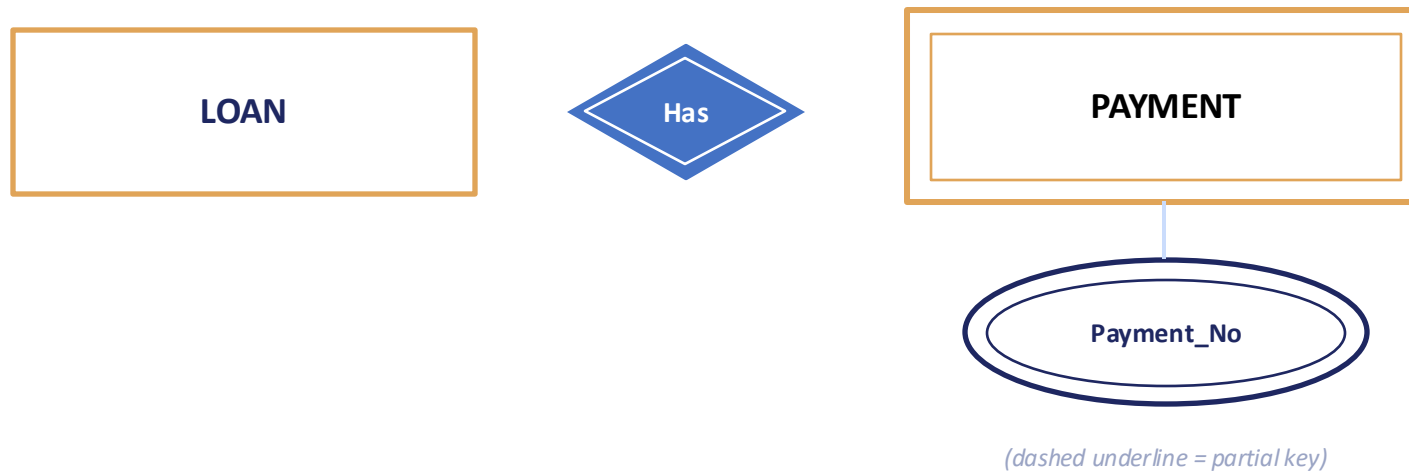
WEAK ENTITY

Definition & Characteristics

A weak entity has no complete primary key of its own; it can only be uniquely identified by combining its partial key with the primary key of its owner (identifying) entity.

- Has only a partial key, called a discriminator — unique only within the scope of one owner
- Cannot exist in the database without a related instance of its owner (identifying) entity
- Participates in an identifying relationship, always total (mandatory) participation on its side
- If the owner entity instance is deleted, all its weak entity instances are deleted too
- Also called a 'dependent entity' or 'subordinate entity'

Diagram



Notation Rules

- Double-line rectangle = weak entity
- Double-line diamond = identifying relationship
- Double-line oval = partial / discriminator key
- Connecting line to owner is often drawn thicker to show total participation

Banking Case Study: LOAN_PAYMENT

Schema Snapshot

LOAN_PAYMENT (Loan_No*, Payment_No, Amount, Date)

Payment_No repeats across different loans (Payment #1 exists for every loan). Only (Loan_No + Payment_No) together is unique.

Owner entity:

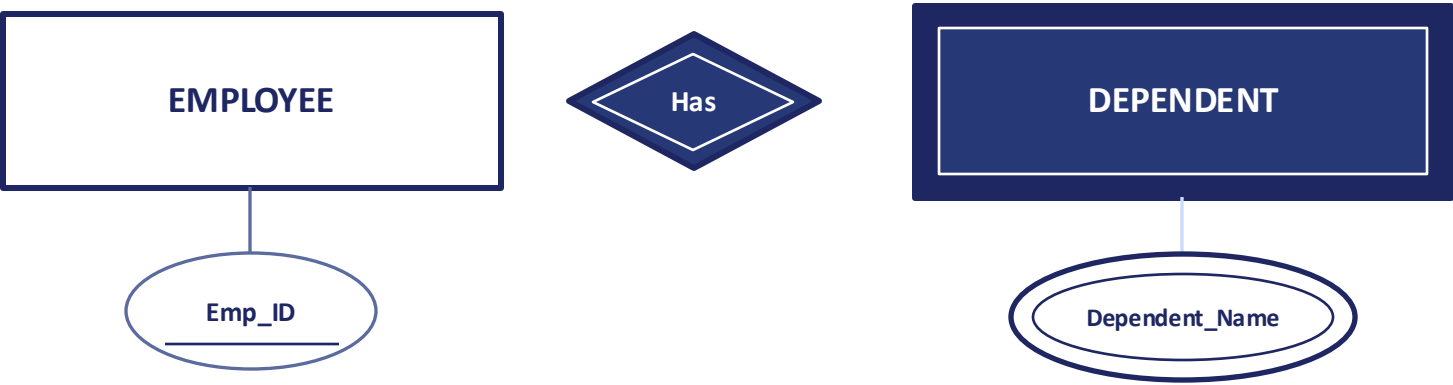
LOAN (Loan_No) — via the identifying relationship 'Has_Payment'

Why It's Weak

- Payment_No alone cannot identify a row — it only makes sense attached to a specific Loan_No
- A LOAN_PAYMENT record cannot be inserted until the corresponding LOAN row already exists
- If a Loan is foreclosed and deleted, all its payment history rows are deleted with it
- In the ER diagram, LOAN_PAYMENT is drawn with a double border and connects to LOAN via a double-diamond relationship

Strong & Weak Entity - Example

HR System: EMPLOYEE (strong) and DEPENDENT (weak)



(dashed underline = partial key, unique only per Employee)

Scenario

An HR system stores each employee's family members as dependents, for medical insurance coverage.

Aspect	EMPLOYEE (Strong)	DEPENDENT (Weak)
Key	Emp_ID — full primary key	Dependent_Name — partial key only
Unique identification	Emp_ID alone identifies the row	Needs Dependent_Name + owner's Emp_ID together
Existence	Exists the moment an employee joins	Cannot exist without a linked Employee record
Deletion effect	Deleting a Dependent doesn't affect Employee	Deleting the Employee deletes all their Dependents

Strong vs Weak Entity Comparison

Aspect	Strong Entity	Weak Entity
Primary key	Owens a complete primary key	Only a partial key (discriminator)
Existence	Independent of other entities	Dependent on its owner entity
ER notation (entity)	Single-bordered rectangle	Double-bordered rectangle
ER notation (key)	Single-bordered oval, underlined	Double-bordered oval, dashed underline
Relationship	Regular relationship (single diamond)	Identifying relationship (double diamond)
Participation	May be partial or total	Always total on the weak entity's side
Deletion effect	Unaffected by other entities' deletion	Deleted automatically if owner is deleted
Examples	EMPLOYEE, STUDENT, ACCOUNT, BOOK	DEPENDENT, NOMINEE, BOOK_COPY, ORDER_ITEM

Real-World Case Studies

BANKING

ACCOUNT

↓ identifies

NOMINEE

A nominee only exists as long as its parent account exists.

HEALTHCARE

PATIENT

↓ identifies

APPOINTMENT

An appointment is meaningless without a specific patient.

LIBRARY

BOOK (title)

↓ identifies

BOOK_COPY

Each copy is identified only by copy number within a book.

E-COMMERCE

ORDER

↓ identifies

ORDER_ITEM

A line item cannot exist without the order it belongs to.

Common Errors When Classifying Entities

Assuming a small or optional-looking entity is automatically weak

Fix: Check for an independent primary key first — size or role in the UI is irrelevant

Forgetting the identifying relationship must have total participation

Fix: The weak entity side must always be mandatory — every instance needs an owner

Using a non-unique attribute as if it were a full primary key

Fix: If it only disambiguates within one owner, it is a partial key/discriminator, not a primary key

Drawing a single-bordered rectangle for a dependent entity

Fix: Weak entities must use a double-bordered rectangle and double diamond in the diagram

Key Takeaways

1

A strong entity owns a full primary key and exists independently of every other entity.

2

A weak entity has only a partial key and depends on an owner entity through an identifying relationship.

3

The distinction directly affects ER notation, cascading deletes, and participation constraints.

4

Classifying entities correctly ensures the schema mirrors real-world business dependencies rather than fighting them.